



ZnO nanorod arrays and direct wire bonding on GaN surfaces for rapid fabrication of antireflective, high-temperature ultraviolet sensors



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ABSTRACT

Rapid, cost-effective, and simple fabrication/package of microscale gallium nitride (GaN) ultraviolet (UV) sensors are demonstrated using zinc oxide nanorod arrays (ZnO NRAs) as an antireflective layer and direct bonding of aluminum wires to the GaN surface. The presence of the ZnO NRAs on the GaN surface significantly reduced the reflectance to less than 1% in the UV and 4% in the visible light region. As a result, the devices fabricated with ZnO NRAs and mechanically stable aluminum bonding wires (pull strength of 3–5 gf) showed higher sensitivity (136.3% at room temperature and 148.2% increase at 250 °C) when compared with devices with bare (uncoated) GaN surfaces. In addition, the devices demonstrated reliable operation at high temperatures up to 300 °C, supporting the feasibility of simple and cost-effective UV sensors operating with higher sensitivity in high-temperature conditions, such as in combustion, downhole, and space exploration applications.

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1. Introduction

Gallium nitride (GaN) has been widely used for high-temperature and high-power-density microelectronics because of its direct, wide bandgap properties [1–3]. In particular, GaN-based sensors have emerged for sensing within harsh environments such as combustion exhausts [4,5], radiation [6,7], and corrosive chemical conditions [8,9] due to their thermal, mechanical, and chemical stability. Among the many types of GaN-based sensors, ultraviolet (UV) photodetectors have especially attracted interest because GaN enables operation within high-radiative and high-temperature environments while absorbing a wide UV photon energy greater than the bandgap of GaN (3.4 eV for Wurtzite crystal structure) [10–14].

The epitaxial growth of GaN films is typically achieved on sapphire and silicon carbide (SiC) substrates using metal organic chemical vapor deposition [15,16], molecular beam epitaxy [17,18], or hydride vapor phase epitaxy [19,20] techniques to minimize lattice mismatch between substrates and grown films [21]. In general, photodetector sensitivity is enhanced using two types of surface modification techniques. One technique leverages surface topography modification (i.e., the shape change of sensor surface),

which increases the sensor surface area and induces multiple collisions of photons on the sensing area [22–24]. In other words, the shape of sensor surface needs to be physically changed by surface roughening or bulk micromachining techniques including photolithography, deposition, and etching processes, which are very challenging processes to implement using sapphire and SiC substrates [25]. The second surface modification architecture uses antireflective surface coatings (films and nanostructures), which have been shown to enhance the sensitivity of optical devices fabricated on planar substrates by reducing the reflectance of the sensing surface without sophisticated microfabrication techniques [26–29]. One well-known antireflective layer is zinc oxide (ZnO), which has been used to enhance devices using thin film or nanowire microstructures [28–34]. ZnO has a direct wide bandgap (~3.3 eV) that is very close to that of GaN and can efficiently absorb the UV photons in the short-wavelength region while simultaneously providing chemical and thermal stability for high-temperature applications [28–32]. Therefore, photodetectors with antireflective layers are able to trap more light and increase absorption of incident photons leading to enhanced optical sensitivity.

To fabricate and package practical optical sensors, metallization is essential for electrical connection between sensor chip and chip carrier. However, device metallization generally includes multiple processes such as photolithography, photoresist development, metal evaporation, and lift-off. In addition, expensive metal sources (e.g., Ti, Al, Pt, Ni, and Au) with specialized equipment (e.g., sputter,

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